Material Safety Data Sheet

MSDS

DAEYANG P&T CO., LTD.

EAGON WHITE Semi-gloss (Exterior) DY#2-WH029

MSDS No.	AA14944-0000000134

1. Identification of the substance/mixture and of the company/undertaking

A. Product name EAGON WHITE Semi-gloss (Exterior) (DY#2-WH029)

B. Recommended use of the product and restrictions on use

- Recommended use: Polyester powder coating for general use
- Restrictions on use: No data available

C. Supplier information (for imports, domestic supplier available for emergency contact)

- Company name: Daeyang P&T Co., Ltd.
- Address: 39-17, Aegibong-ro 774beon-gil, Haseong-myeon, Gimpo-si, Gyeonggi-do, Korea
- Emergency phone number: 031) 987-8587

2. Hazards identification

A. Hazard classification

- Acute toxicity (oral) : Category 4
- Skin corrosion/irritation : Category 2
- Carcinogenicity : Category 2

B. Label elements, including precautionary statements

- Pictograms



- Signal word: Warning
- Hazard statements:
  H351 Suspected of causing cancer
  H315 Causes skin irritation
  H302 Harmful if swallowed
- Precautionary statements
  - Prevention:
    P201 Obtain special instructions before use.
    P202 Do not handle until all safety precautions have been read and understood.
    P264 Wash hands thoroughly after handling.
    P270 Do not eat, drink or smoke when using this product.
    P280 Wear protective gloves/protective clothing/eye protection/face protection.
  - Response:
    P301+P312 IF SWALLOWED: Call a POISON CENTER/doctor if you feel unwell.
    P302+P352 IF ON SKIN: Wash with plenty of water.
    P308+P313 IF exposed or concerned: Get medical advice/attention.
    P321 Specific treatment (see on this label).
    P330 Rinse mouth.
    P332+P313 If skin irritation occurs: Get medical advice/attention.
    P362+P364 Take off contaminated clothing and wash it before reuse.
  - Storage: P405 Store locked up.
  - Disposal: P501 Dispose of contents/container in accordance with waste regulations.

3. Composition / information on ingredients

Chemical name	Common name / Synonyms	CAS No.	Content (%)
Polyester resin	Polymer of 1,3-benzenedicarboxylic acid with 1,4-benzenedicarboxylic acid, 2,2-dimethyl-1,3-propanediol, 1,2-ethanediol and hexanedioic acid	40471-09-8	55 ~ 59
N,N,N',N'-Tetrakis(2-hydroxyethyl)adipamide	-	6334-25-4	1 ~ 5
Polybutyl acrylate	2-Propenoic acid, butyl ester, homopolymer (2-PROPENOIC ACID, BUTYL ESTER,	9003-49-0	0.83 ~ 0.87
Benzoin	Ethanone, 2-hydroxy-1,2-diphenyl- (ETHANONE, 2-HYDROXY-1,2-DIPHENYL);	119-53-9	0.1 ~ 0.5
Acrylic acid-ethylene copolymer	2-Propenoic acid, polymer with ethene (2-PROPENOIC ACID, POLYMER WITH ETHENE);	9010-77-9	3.2 ~ 3.6
3,9-Bis(2,4-bis(1,1-dimethylethyl)phenoxy)-2,4,8,10-tetraoxa-3,9-diphosphaspiro[5.5]undecane	SONOX626	26741-53-7	0.1 ~ 0.4
Aluminium hydroxide	-	21645-51-2	7.54 ~ 7.58
Limestone	-	1317-65-3	4.6 ~ 5.0
Titanium dioxide	-	13463-67-7	22.3 ~ 22.7
Carbon black	-	1333-86-4	0.01 ~ 0.04
C.I. Pigment Yellow 83	2,2'-((3,3'-Dichloro(1,1'-biphenyl)-4,4'-diyl)bis(azo))bis(N-	5567-15-7	0.11 ~ 0.15
Iron oxide	-	1309-37-1	0.03 ~ 0.07

4. First-aid measures

A. Eye contact

- Get emergency medical attention
- In case of contact, immediately flush skin and eyes with running water for at least 20 minutes

B. Skin contact

- If you feel unwell, get medical advice/attention.
- Remove contaminated clothing and shoes and isolate the contaminated area
- In case of contact, immediately flush skin and eyes with running water for at least 20 minutes
- In case of minor skin contact, prevent spreading of contamination

C. Inhalation

- IF exposed or concerned: Get medical advice/attention.
- Move to fresh air
- Keep warm and at rest

D. Ingestion

- IF exposed or concerned: Get medical advice/attention.
- If the substance has been swallowed or inhaled, do not give mouth-to-mouth resuscitation; use appropriate respiratory medical equipment

E. Notes to physician

- In case of exposure, contact medical staff and take special first-aid measures such as follow-up examination.
- Ensure that medical personnel are aware of the material involved and take precautions to protect themselves

5. Fire-fighting measures

A. Suitable (unsuitable) extinguishing media

○ Suitable (unsuitable) extinguishing media	For fires involving this material, use alcohol-resistant foam, carbon dioxide or water spray For smothering, use dry sand or earth

B. Specific hazards arising from the chemical

○ Specific hazards arising from the chemical	May decompose at high temperature to produce toxic gases Containers may explode when heated Some may burn but do not ignite readily Non-flammable; the substance itself does not burn but may decompose on heating to produce corrosive/toxic fumes

C. Special protective equipment and precautions for firefighters

- Rescuers should wear appropriate protective equipment.
- Fight fire from a safe distance outside the area

6. Accidental release measures

A. Personal precautions and protective equipment

- Do not breathe dust/fume/gas/mist/vapours/spray.
- Wipe up spills immediately and follow the precautions in the protective equipment section.
- Isolate the contaminated area.
- Keep unnecessary and unprotected personnel from entering.
- Remove all sources of ignition
- Stop leak if it can be done without risk
- Do not touch damaged containers or spilled material unless wearing appropriate protective clothing
- Cover with plastic sheet to prevent spreading
- Note materials and conditions to avoid

B. Environmental precautions

Prevent entry into waterways, sewers, basements or confined areas

C. Methods and materials for containment and cleaning up

- Absorb spills with inert material (e.g. dry sand or earth) and place in a chemical waste container.
- Absorb liquid and wash the contaminated area with detergent and water.

7. Handling and storage

A. Precautions for safe handling

- Do not handle until all safety precautions have been read and understood.
- Follow all MSDS/label precautions even after the container is emptied, as it may retain product residue.
- Use with care during handling/storage.
- Open the lid carefully before opening.
- Do not enter storage areas without adequate ventilation.
- Note materials and conditions to avoid

B. Conditions for safe storage

- Store locked up.

- Drain empty drums completely, close them properly and return them promptly to the drum reconditioner or dispose of them appropriately.

8. Exposure controls / personal protection

A. Occupational exposure limits, biological exposure limits, etc.

○ Korea

Titanium dioxide	TWA - 10 mg/m3 (Carcinogen category 2)
Limestone	TWA - 10 mg/m3 (Limestone)
Aluminium hydroxide	TWA - 10 mg/m3 (Aluminium metal dust)
Iron oxide	TWA - 5mg/m3
Carbon black	TWA - 3.5 mg/m3 (Carcinogen category 2, inhalable)

○ ACGIH

Titanium dioxide	TWA - 10 mg/m^3
Iron oxide	TWA - 5 mg/m^3 (respirable)
Aluminium hydroxide	TWA - 1 mg/m^3
Carbon black	TWA - 3 mg/m^3

B. Appropriate engineering controls

Use process enclosure, local exhaust ventilation or other engineering controls to keep airborne levels below exposure limits.

C. Personal protective equipment

- Wear respiratory protection certified by the Korea Occupational Safety and Health Agency suitable for the physicochemical properties of the particulates to which you are exposed
- At concentrations below 100 mg/m3, wear a half-mask respirator with an appropriate type of filter
- At concentrations below 250 mg/m3, wear a loose-fitting hood/helmet powered air-purifying respirator or continuous-flow dust mask with an appropriate type of filter
- At concentrations below 500 mg/m3, wear a full-face or powered half-mask, or supplied-air continuous-flow/pressure-demand half-mask respirator with appropriate filter
- At concentrations below 10000 mg/m3, wear a full-face or helmet/hood type pressure-demand supplied-air respirator with appropriate filter
- At concentrations below 100000 mg/m3, wear a self-contained breathing apparatus (SCBA) or pressure-demand SCBA with appropriate filter
- In case of oxygen deficiency (<19.6%), wear a supplied-air respirator or self-contained breathing apparatus
- For particulates, the following respiratory protection is recommended
- - Filtering facepiece dust mask or air-purifying dust mask (high-efficiency particulate filter), or powered fan dust mask (filter for dust, mist and fume)
- - Filtering facepiece dust mask or air-purifying dust mask (high-efficiency particulate filter), or powered fan dust mask (filter for dust, mist and fume)

9. Physical and chemical properties

A. Appearance

○ Physical state	Powder
○ Colour	White
B. Odour	Odourless
C. Odour threshold	No data available
D. pH	No data available
E. Melting point/freezing point	No data available

F. Initial boiling point and boiling range	No data available
G. Flash point	No data available
H. Evaporation rate	No data available
I. Flammability (solid, gas)	No data available
J. Upper/lower flammability or explosive limits	No data available
K. Vapour pressure	No data available
L. Solubility	No data available
M. Vapour density	No data available
N. Relative density	No data available
O. n-Octanol/water partition coefficient (Kow)	No data available
P. Auto-ignition temperature	No data available
Q. Decomposition temperature	No data available
R. Viscosity	No data available
S. Molecular weight	No data available

10. Stability and reactivity

A. Chemical stability and possibility of hazardous reactions

- Inhalation of the substance may be harmful
- Stable under recommended storage and handling conditions
- Hazardous polymerization will not occur

11. Toxicological information

A. Information on likely routes of exposure

- Short-term exposure causes irritation, nausea, vomiting, headache and narcosis
- No information on serious adverse effects of short-term exposure
- Short- and long-term exposure causes irritation
- Short-term exposure causes irritation
- Can be absorbed into the body by inhalation
- Can be absorbed into the body by inhalation and ingestion
- Can be absorbed into the body through the skin, digestive tract and by inhalation of aerosol
- Can be absorbed into the body by inhalation of vapour
- Can be absorbed into the body by inhalation, skin contact and ingestion

B. Information on health hazards

○ Acute toxicity

- Oral

Benzoin	LD50 6400 mg/kg Rat
Titanium dioxide	LD50 > 2000 mg/kg Mouse (OECD TG 420)
Aluminium hydroxide	LD50 > 2000 mg/kg Rat (OECD TG 423, GLP)
SONOX626	LD50 > 5500 mg/kg Rat
Iron oxide	LD50 > 10000 mg/kg Rat
Carbon black	LD50 > 8000 mg/kg Rat (OECD TG 401)
C.I. Pigment Yellow 83	LD50 > 5000 mg/kg Rat

- Dermal
  - Benzoin: LD50 8870 mg/kg Rabbit
  - SONOX626: LD50 ≥ 2000 mg/kg Rabbit
  - Carbon black: LD50 > 8000 mg/kg Rabbit
- Inhalation
  - Titanium dioxide: Dust LC50 > 6.82 mg/ℓ Rat (OECD TG 403, no deaths)
  - Iron oxide: Dust LC50 5.05 mg/ℓ 4 hr Rat (OECD TG 403, GLP)
  - Carbon black: Dust LC50 > 4.6 mg/m^3 4 hr Rat (no deaths)
- Skin corrosion or irritation
  - Benzoin: Rabbit, no effect, STANDARD DRAIZE TEST, Not Reported
  - Titanium dioxide: Rabbit skin corrosion/irritation test: not irritating, erythema score = 0, OECD TG 404
  - Aluminium hydroxide: Not irritating, Rabbit, OECD TG 404
  - Iron oxide: Rabbit skin irritation test (OECD TG 404): not irritating
  - Acrylic acid-ethylene copolymer: Causes skin irritation
  - Carbon black: Rabbit skin irritation test (OECD TG 404): not irritating
- Serious eye damage or eye irritation
  - Titanium dioxide: Rabbit serious eye damage/irritation test: not irritating, conjunctival redness score = 1-2, OECD TG 405, GLP
  - Aluminium hydroxide: Not irritating, Rabbit, fully reversible within 48 hours, OECD TG 405
  - Iron oxide: Rabbit eye irritation test (OECD TG 405, GLP): not irritating
  - Acrylic acid-ethylene copolymer: Causes eye irritation
  - Carbon black: Rabbit eye irritation test (OECD TG 405): not irritating
- Respiratory sensitisation: No data available
- Skin sensitisation
  - Titanium dioxide: Guinea pig skin sensitisation test: no skin sensitisation, OECD TG 403
  - Aluminium hydroxide: Not sensitising, Guinea pig, Guinea Pig Maximisation Test (GPMT), OECD TG 406
  - Iron oxide: Guinea pig sensitisation test: not sensitising
  - Carbon black: Guinea pig skin sensitisation test (OECD TG 406, GLP): not sensitising
- Carcinogenicity
  - Occupational Safety and Health Act: No data available
- Ministry of Employment and Labor notice
  - Titanium dioxide: 2
  - Carbon black: 2
- IARC
  - Titanium dioxide: 2B
  - Iron oxide: 3
  - Carbon black: 2B
- OSHA
  - Carbon black: Applicable
- ACGIH
  - Titanium dioxide: A4

Iron oxide	A4
Carbon black	A3
- NTP	No data available
- EU CLP	
Titanium dioxide	2 (only for powder containing 1 % or more of particles with aerodynamic diameter of 10 μm or less)
- Ministry of Environment	No data available
- NITE	No data available
- Germ cell mutagenicity	
Titanium dioxide	In vitro bacterial reverse mutation test OECD TG 471, mammalian cell gene mutation test OECD TG 476 and chromosomal aberration test OECD TG 473: negative with and without metabolic activation; in vivo chromosomal aberration and micronucleus tests: negative
Aluminium hydroxide	In vitro chromosomal aberration test in mammalian cells: positive (without metabolic activation), OECD TG 473
Iron oxide	Negative in reverse mutation test, chromosomal aberration test (OECD TG 473, GLP) and in vivo comet assay
SONOX626	Negative (reverse mutation test, chromosomal aberration test)
Carbon black	Negative in bacterial reverse mutation, gene mutation, sister chromatid exchange and Drosophila tests (OECD TG 471, 476, 477, 479)
- Reproductive toxicity	
Titanium dioxide	Reproductive/developmental toxicity test in rats: no effects on clinical signs, body weight, etc. NOAEL = 1000 mg/kg bw/day (OECD TG 210)
Carbon black	Teratogenicity/developmental toxicity test in rats: no reproductive toxicity (OECD TG 414)
- Specific target organ toxicity (single exposure)	
Titanium dioxide	Acute oral toxicity test in rats: no deaths, no body weight changes and no significant lesions at necropsy, OECD TG 425
Acrylic acid-ethylene copolymer	Irritates the respiratory tract on inhalation
Iron oxide	Causes respiratory tract irritation
- Specific target organ toxicity (repeated exposure)	
Titanium dioxide	NITE classification 2
Iron oxide	Repeated inhalation exposure may affect the respiratory tract (siderosis)
Carbon black	Repeated exposure over 10 years or more causes chronic bronchitis, impaired lung function, pneumoconiosis, etc.
- Aspiration hazard	No data available
○ Other hazardous effects	No data available

12. Ecological information

A. Ecotoxicity

- Fish	
Benzoin	LC50 226.385 mg/ℓ 96 hr
Titanium dioxide	LC50 > 100 mg/ℓ 96 hr Carassius auratus (OECD Guideline 203)
Aluminium hydroxide	NOEC > 50 mg/ℓ 96 hr Ictalurus punctatus
Iron oxide	LC0 ≥ 50000 mg/ℓ 96 hr Danio rerio
SONOX626	LC50 100 mg/ℓ 96 hr

Carbon black	LC50 > 1000 mg/ℓ 96 hr Tribolodon hakonensis

- Crustaceans

Benzoin	LC50 20.569 mg/ℓ 48 hr
Titanium dioxide	LC50 > 500 mg/ℓ 48 hr Daphnia magna
Aluminium hydroxide	NOEC > 22.6 mg/ℓ 96 hr Acronuria sp.
Iron oxide	EC50 > 100 mg/ℓ 48 hr
SONOX626	EC50 92.2 mg/ℓ 48 hr Daphnia magna
N,N,N',N'-Tetrakis(2-hydroxyethyl)adipamide	LC50 24000 mg/ℓ 48 hr (estimated)
Carbon black	EC50 > 5600 mg/ℓ 24 hr Daphnia magna

- Algae

Titanium dioxide	EC50 > 50 mg/ℓ 72 hr Selenastrum capricornutum
N,N,N',N'-Tetrakis(2-hydroxyethyl)adipamide	EC50 29.401 mg/ℓ 96 hr (estimated)
Carbon black	ErC50 > 10000 mg/ℓ 72 hr Desmodesmus subspicatus

B. Persistence and degradability

- Persistence

Benzoin	log Kow 1.84 (estimated)
SONOX626	log Kow 4.95 (25 °C)
N,N,N',N'-Tetrakis(2-hydroxyethyl)adipamide	log Kow -3.82 (estimated)

- Degradability — No data available

C. Bioaccumulative potential

- Bioaccumulation — No data available
- Biodegradability — No data available

D. Mobility in soil — No data available

E. Other adverse effects — No data available

13. Disposal considerations

A. Disposal methods

Where two or more designated wastes are mixed and difficult to separate, they may be reduced and stabilized by incineration or a similar method

B. Precautions for disposal

Dispose of contents/container in accordance with waste regulations

14. Transport information

A. UN number	No UN transport classification information
B. Proper shipping name	Not applicable
C. Transport hazard class	Not applicable
D. Packing group	Not applicable
E. Marine pollutant	No data available

F. Special precautions for users related to transport or means of transport

○ Emergency measures in case of fire	No data available
○ Emergency measures in case of spillage	No data available

15. Regulatory information

A. Regulations under the Occupational Safety and Health Act

Limestone	Substance subject to work environment monitoring (every 6 months) Substance subject to special health examination (every 24 months) Substance with occupational exposure limit
Titanium dioxide	Hazardous substance subject to control Substance subject to work environment monitoring (every 6 months) Substance with occupational exposure limit
Aluminium hydroxide	Hazardous substance subject to control Substance subject to work environment monitoring (every 6 months) Substance subject to special health examination (every 12 months) Substance with occupational exposure limit
Iron oxide	Hazardous substance subject to control Substance subject to work environment monitoring (every 6 months) Substance subject to special health examination (every 12 months) Substance with occupational exposure limit
Carbon black	Substance with occupational exposure limit

B. Regulations under the Chemical Substances Control Act	Not applicable
C. Regulations under the Act on Registration and Evaluation of Chemicals	Not applicable

D. Regulations under the Dangerous Substances Safety Management Act

Benzoin	Designated waste
Aluminium hydroxide	Designated waste
Iron oxide	Designated waste
SONOX626	Designated waste

F. Other domestic and foreign regulations

○ Domestic regulations	
- Other domestic regulations	Not applicable
○ Foreign regulations	
- US regulations (OSHA)	Not applicable
- US regulations (CERCLA)	Not applicable
- US regulations (EPCRA 302)	Not applicable
- US regulations (EPCRA 313)	Not applicable
- US regulations (Rotterdam Convention substances)	Not applicable
- US regulations (Stockholm Convention substances)	Not applicable

- US regulations (Montreal Protocol substances) Not applicable
- EU classification (harmonised classification) Not applicable
- EU classification (risk phrases) Not applicable
- EU classification (safety phrases) Not applicable

16. Other information

A. Sources of data

- Korea Chemical Information System
- Prepared based on KOSHA, NITE, ESIS, NLM, SIDS, IPCS, etc.

B. Date of first issue 2024-05-09

C. Number of revisions and date of last revision

○ Number of revisions 1

○ Date of last revision 2026-01-06

D. Other

※ This Material Safety Data Sheet (MSDS) has been edited and partly modified with reference to the MSDS provided by the Korea Occupational Safety and Health Agency; commercial use is prohibited